Ghana Farm Intelligence

Farm report - Sample farm - Bolgatanga area (Upper East)

Generated 31 Oct 2026 16:57 UTC. Africa/Accra (GMT). All areas are geodesic (WGS84).

Area	25.03 ha (61.86 acres)
Perimeter	1,929 m
Region / district	Upper East Region / Bolgatanga Municipal
Agro-ecological zone	Sudan Savannah
Crop	Maize
Planting date	-
Irrigation	rainfed

Recommended next actions

1. Protect soil moisture over the next week

Mulch around plants where material is available, weed early to reduce competition, and irrigate only if you have water and the crop is at a critical stage.

Why: The forecast shows a 16-day dry spell with a water balance of -34.7 mm (rainfall minus reference evapotranspiration) over the next 7 days.

2. Build soil organic matter

Keep crop residues on the field where possible, apply well-rotted manure or compost, and consider a legume in rotation. Avoid burning residues.

Why: Estimated soil organic carbon is 6.8 g/kg, which is low for sustained cropping.

Soil (SoilGrids estimates, 250 m)

Property	Value	Unit	Valid coverage
pH (in water)	6.24	pH	100%
Soil organic carbon	6.77	g/kg	100%
Total nitrogen	0.71	g/kg	100%
Clay	26.05	%	100%
Sand	53.43	%	100%
Silt	20.52	%	100%
Cation exchange capacity	11.26	cmol(c)/kg	100%
Bulk density	1.56	kg/dm3	100%
Coarse fragments	1.01	vol %	100%

Texture class: Sandy clay loam. These are model predictions, not laboratory measurements. A local soil test is required before lime or fertiliser prescriptions.

Weather

Current: 32.7 deg C, 52.0% humidity, Mainly clear, wind 7.0 km/h. Model grid cell at 10.790861, -0.954895 (~11 km).

Date	Rain (mm)	Prob.	Tmax	Tmin	ET0
2026-10-31	0.0	0.0	35.2	25.0	4.87
2026-11-01	0.0	0.0	34.9	23.8	4.94
2026-11-02	0.0	4.0	36.0	24.3	5.03
2026-11-03	0.0	12.0	35.4	25.6	5.0
2026-11-04	0.0	2.0	35.1	25.7	4.58
2026-11-05	0.0	0.0	36.8	23.7	5.57
2026-11-06	0.0	0.0	35.7	22.5	4.68

Rainfall (CHIRPS)

Period	Observed	1991-2020 baseline	Percent of normal
Last 7 days (2026-09-24 to 2026-09-30)	34.9 mm	42.0 mm	83.0%

Period	Observed	1991-2020 baseline	Percent of normal
Last 30 days (2026-09-01 to 2026-09-30)	205.6 mm	179.9 mm	114.0%
Last 90 days (2026-07-03 to 2026-09-30)	585.5 mm	613.3 mm	95.0%

Latest CHIRPS day available: 2026-09-30 (31 days before today). ~5.5 km estimate, not a rain gauge.

Satellite vegetation (Sentinel-2)

Latest usable image: 2026-10-29 (scene S2A_30PYS_20261029_0_L2A). Cloud-free over the field: 100%. NDVI mean 0.3136, NDMI mean -0.177. Indices show relative condition and are not a pest, disease or nutrient diagnosis.

Crop suitability (rule-based screening)

Crop	Category	Score	Confidence	Perennial
Cassava	Highly suitable	100.0	high	no
Maize	Highly suitable	95.0	high	no
Upland rice	Highly suitable	95.0	high	no

Screening only. No yield, price or economic score is produced. See the app for the deciding factors, missing inputs and cited crop ranges.

Data sources and licences

Dataset	Provider	Licence	Resolution	Status
Open-Meteo forecast (best_match global model)	openmeteo_forecast	CC BY 4.0 (data); hosted free tier is non-commercial	~11 km model grid	ok
ISRIC SoilGrids 2.0 - pH, SOC, nitrogen, texture, CEC, bulk density, coarse fragments	soilgrids	CC BY 4.0	250 m (native equal-area grid)	ok
ESA WorldCover 2021 v200 (10 m)	esa_worldcover	CC BY 4.0 - © ESA WorldCover project 2021 / Contains modified Copernicus Sentinel data (2021)	10 m	ok
Copernicus DEM GLO-30 (30 m)	copernicus_dem	Copernicus DEM - © DLR e.V. 2010-2014 and © Airbus DS 2014-2018, provided under COPERNICUS by the EU and ESA	30 m (1 arc-second)	ok
CHIRPS v2.0 daily and monthly rainfall (Digital Earth Africa mirror)	chirps	CHIRPS: public domain (UCSB CHC); DE Africa mirror: CC BY 4.0	0.05 deg (~5.5 km)	ok
CHIRPS 1991-2020 monthly climatology (Ghana subset)	chirps	CHIRPS: public domain (UCSB CHC)	0.05 deg (~5.5 km)	ok
Copernicus Sentinel-2 L2A (10 m/20 m) - AWS Open Data COGs via Element84 STAC	sentinel2	Copernicus Sentinel data - free and open (contains modified Copernicus Sentinel data)	10 m (red/NIR), 20 m (SWIR, SCL resampled to 10 m)	ok
FAO WaPOR v3 - AETI, reference ET, net primary production	wapor	CC BY-NC-SA 4.0	100 m / 300 m	unavailable
Ghana crop reference database (FAO EcoCrop, FAO crop calendar, FAO-56, MoFA, GIEWS)	ghana_crop_reference	Compiled reference data; each source's own licence applies (see attribution page)	Crop-level reference ranges	ok

Remote sensing and modelled data cannot replace soil testing, field inspection or extension advice. Ghana Farm Intelligence does not infer land ownership and does not make definitive pest, disease, fertiliser or yield claims from imagery. Farm boundaries are private and are not published.